Candidate Number (or Examination Number) ____________________ Fraction ____________________

The **32nd** Republic of China Biology Olympiad National Team Selection Camp , **2021**

Animal Physiology and Marine Biotechnology

Practice **A**

I. Materials and Experimental Equipment:

Experimental equipment	quantity	Medicines and materials	quantity
1. Photopolymerization system		1. Agarose film (in the electrophoresis tank) for common use	1 piece
2. Mini horizontal electrophoresis system	Group 1, Item	2. Numbered petri dishes (for holding film)	1
3. Micro-dispenser p10	One	3. 100bp nucleic acid standard (M) (place on ice)	1 tube (10μl)
4. 10 μl micropipette tip	20 vials	4. DNA fluorescent nucleic acid dye (place on ice)	1 tube (10μl)
5. Timer	1	5. ARMS-PCR product 1 (1) (placed on ice)	1 tube (10μl)
6. Scissors	1	6. ARMS-PCR product 2 (2) (placed on ice)	1 tube (10μl)
7. Glue	1	7. ARMS-PCR product 3 (3) (placed on ice)	1 tube (10μl)

Please note:

1. The total time allotted for this exam, including both Practice A and Practice B, is 90 minutes.
2. The materials and equipment on the table will not be replenished once they are used up.
3. This exam paper (including the cover and the question paper) consists of 6 pages and must be returned in its entirety upon submission.
4. Please answer the questions on this paper. You may write your answers on the back, but please indicate the question number.

«Exam Number» Colloidal electrophoresis analysis of amplification refractory mutagenesis system polymerase chain reaction **(PCR)**

Polymerase Chain (ARMS-PCR) product

I. Experiment Objective:

ARMS-PCR products were analyzed by colloidal electrophoresis to detect single nucleotide polymorphism (SNP) genotypes.

II. Introduction to the Experimental Background:

Various colors of leopard chub (Plectropomus leopardus) are commonly found on Taiwan's coral reefs and coastal areas. Researchers, referencing literature, discovered a strong correlation between the SNP genotype carried by a specific DNA sequence regulating skin color variation and different leopard chub colors. Based on the literature, researchers hypothesize that the SNP genotype for red-skinned leopard chub should be [A/A], and the SNP genotype for red-gray-skinned leopard chub should be [...]. The SNP genotype of the gray-skinned leopard shad should be [A/C], and the genotype should be [C/C]. Researchers collected several leopard shad with different body colors from the coastal areas of Taiwan and decided to use ARMS-PCR to determine the genotypes associated with body color. ARMS-PCR, introduced in 1989, is a method for SNP genotyping. This method utilizes the lack of 3' to 5' exonuclease activity in Taq DNA polymerase, which hinders the amplification rate of DNA products when base pairing errors occur. ARMS-PCR requires four primers: two outer primers and two inner primers. The two inner primers contain 3' end variants to specifically amplify specific products, which are then analyzed by gel electrophoresis to detect SNP genotypes. This experiment will perform ARMS-PCR on DNA sequences associated with different skin color traits to facilitate subsequent analysis of SNP molecular marker genotypes related to different body color traits.

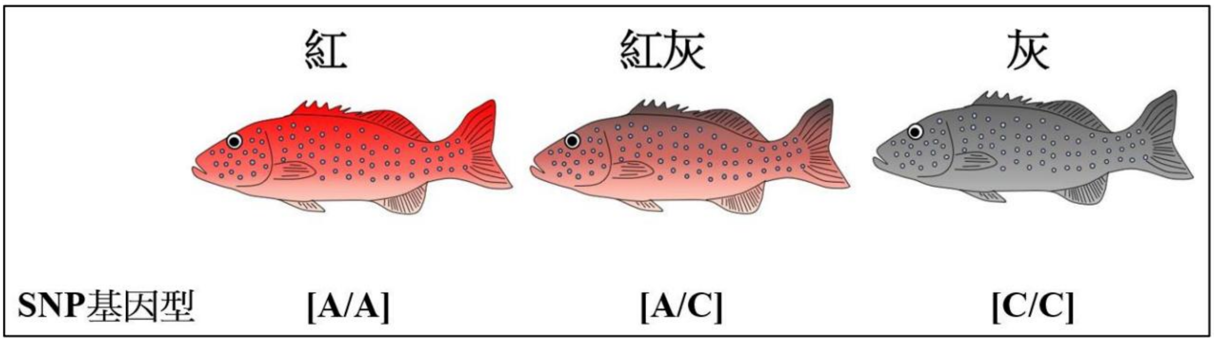

Figure 1. Researchers inferred the SNP genotypes carried by different skin colors in the leopard bream (Plectropomus leopardus) based on literature.

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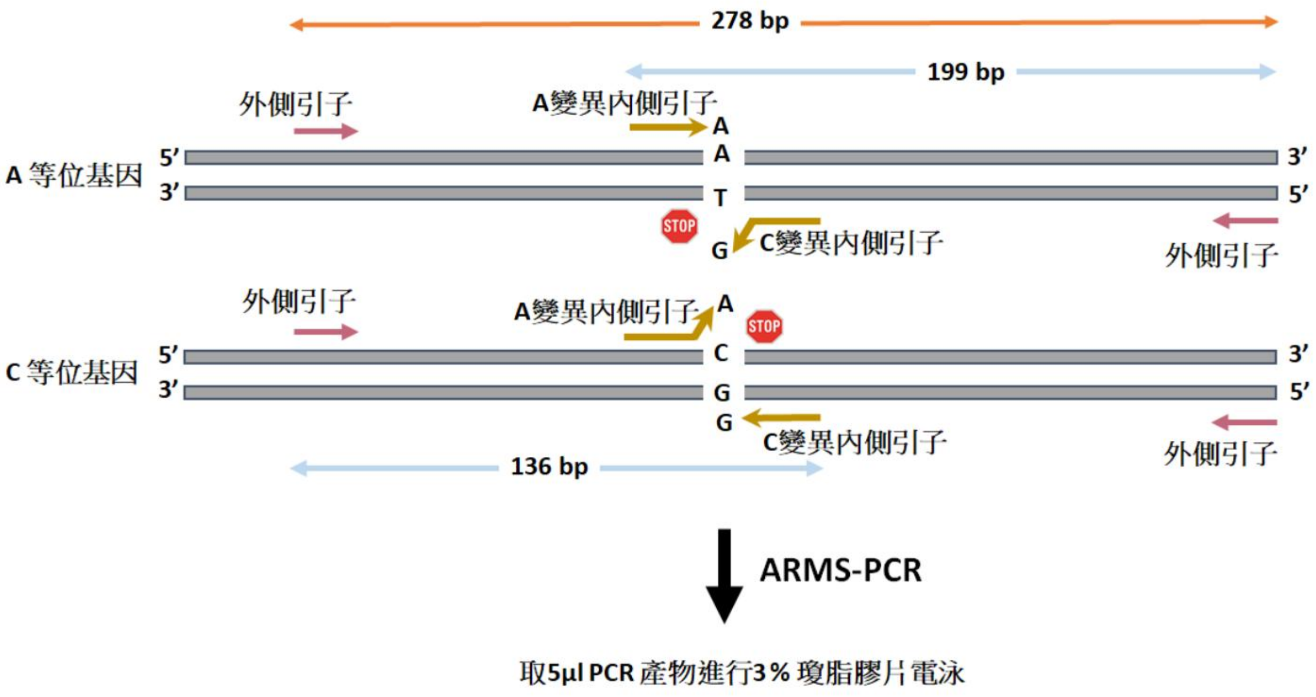


Figure 2. Schematic diagram of ARMS-PCR principle shows the length of the target DNA sequence that can be extracted by the inner and outer primers.

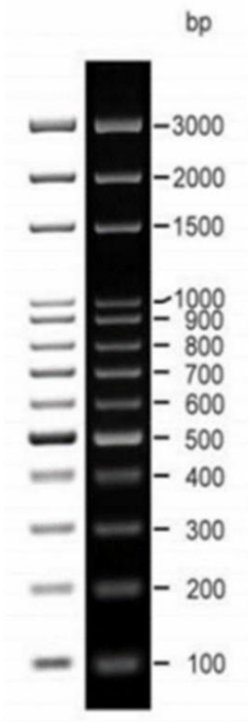


Figure 3. Standard diagram of DNA separation mode used in this experiment.

IV. Experimental Procedure:

1. Add 2 μl of DNA fluorescent dye to the three tubes of ARMS-PCR products (1, 2, 3) and nucleic acid standard (M), and dilute by micro-dosage.

   Mix thoroughly with a syringe and let stand for 1 minute.

2. Using a microdispenser, take 10 μl of the ARMS-PCR product sample mixed with DNA fluorescent dye and nucleic acid standards, from left to right.

   Nucleic acid standard, sample 1, sample 2, and sample 3 are injected sequentially into the sample slot of the electrophoresis film.

3. Cover the electrophoresis tank and connect the wires to the tank (red (+) pole, black (-) pole), turn on the switch, and apply 100 volts.

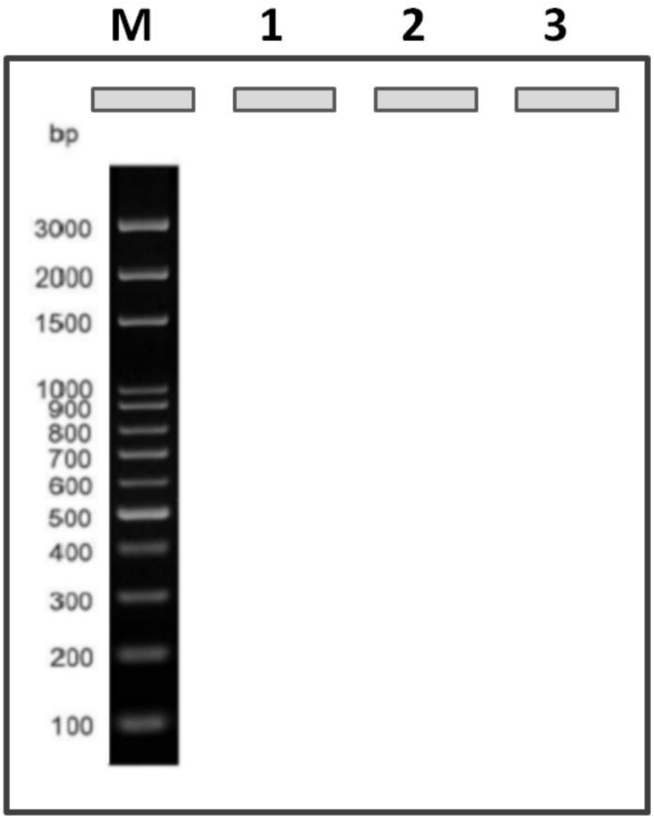
   After about 40 minutes of electrophoresis, the yellow dye will be visible moving to the bottom of the electrophoresis gel. At this point, the power can be turned off.

4. Take out the electrophoresis gel, place it in a numbered culture dish, and take it to the common area to have the teaching assistant take a picture using the gel imaging system. Print out the results.

5. Please cut out the electrophoresis results and paste them below:

V. Exam Questions:

(1) Please mark the positions of the **PCR** products of samples **1 ,** 2 , and **3** on the graph below. (9%)



(2) What colors could the fish in samples **1 ,** 2 , and **3** possibly be? (9%)

(3) Following on to question two, please explain your reasoning for judging **the three** fish skin colors. (5%)

«Exam Number» **(4)** Why **is agarose used instead of agar when** making **DNA** films? (5%)

(5) Try to deduce why the content of magnesium chloride ($MgCl_2$) needs to be optimized in **the ARMS-PCR experiment. What is the purpose? (4%)**

(6) Researchers discovered a SNP in a **DNA** sequence with the genotype [C/G], which is significantly related to the body color of leopard shad. Please design an **18** -base inner primer based on the following sequence to facilitate the researchers' genotype determination of this **SNP . (18%)**

外側引子

5'- **ATACGATTTTAATGTGTC**ATTCAGACCTTGAATTACATCATGGTCCAGGTTGTCATAAA
GAGCAACCACTATGACAGGCTTTCCACGCCTCCAACCTCCCTCTGTCCCCACCTCCAA
GATCCAACAGCCAATCAGAGGAGACCTCTGCTGCAGCACTGGGCTATAAAAACCTCAC
AATGCTGTTCATCAGCACACATCAGAGACAGAGAGATCAGAAGCAGCTTTCTGACTAA
AACAGACTCTCAGCAGACAAAAAGATGAAGATGCCTCAGCTACTCATCCTCCTGCTGG
CTATTCTGCTGCACGCAGCAGCCG**[C/G]**TTTCCCTGCAGCTCTGCCTGGCCGAGACA
AACACGCCTCCTGGGACGACGTGAACGTGGTGGCTCATGGGCTCCTGCAGCTGGGCC
AGGGTCTGAAGGAGCATGTGGACAAGACCAAAATCCAGATGAGCGACGTCAGTGCCA
AACTGAAGGCCCTCAATGGCACAGTGGCCGAACTGGAGAGGAAGCAGCGGGAGCAA
AATGAAGCTCTGAAGGCAAGAGGCAAAGAGGAGGA**GGAGGAGGAGAGGGAGA** -3'

外側引子

2021 Selection Camp Practical Exam Number:

Candidate Number ____________________ Fraction ____________________

The **32nd** Republic of China Biology Olympiad National Team Selection Camp , **2021**

Animal Physiology and Marine Biotechnology

Practice **B**

Materials and experimental equipment:

Equipment and Quantity	Quantity of equipment and quantity		quantity
1. Place the clawed toad tadpoles in a petri dish. (Post the exam number)	5 x 9.3mL droppers (reusable) 1 tube		
2. Place the bullfrog tadpoles in a petri dish. (Post the exam number)	Five 10-legged toad tadpoles were anesthetically treated.		1 plate
3. Transparent inner box (containing a suitable amount of water)	1. Anesthetic solution for bullfrog tadpoles.		1 plate
4. Black/white outer box	One 12. P10 pipetteman (micropipette)	(For practical use	1 branch
5. Dust-free paper	only, use A) 1 box of 13. P10 straw tips	(Practice A is shared)	1 box
6. Extremely fine needle	1. Timer (Do not press for timing purposes)		1
7.15 cm ruler	1 dissecting microscope (model 15)		1 unit
8. Tweezers	1 piece		

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2. The materials and equipment on the table will not be replenished once they are used up.
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4. Please answer the questions on this paper. You may write your answers on the back, but please indicate the question number.

An Exploration of Tadpoles' Response to Environmental Brightness and Their Visual Role

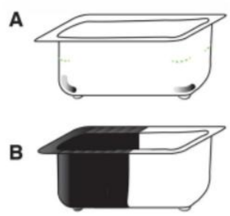
(A simple behavioral assay for testing visual function in tadpoles)

I. Introduction

Ecology is the study of the interaction between organisms and their environment. The environment broadly refers to both abiotic factors (temperature, humidity, pH, heavy metal pollutants, etc.) and biotic factors (predators, food, competitors, etc.) encountered by specific organisms. This experiment uses tadpoles of *bullfrogs (Rana catesbeiana) at 12 days* old and *clawed frogs (Xenopus* laevis) at 14 days old as materials to explore behavioral changes in their response to environmental brightness preferences. Since tadpoles can also be used to study the correlation between the development of the visual nervous system at different stages and the formation of vision, this experiment will explore the correlation between tadpoles' visual responses to external light.

Materials and Equipment

1. Place 5 clawed frog tadpoles in a petri dish (label with exam number). 2. Place 5 bullfrog tadpoles in a petri dish (label with exam number).



3. One transparent inner box (containing an appropriate amount of water), as shown in Figure A on the right.
4. One black/white outer box (black represents a dark environment, white represents a bright environment, as shown in Figure B on the right).
5. One box of dust-free paper
6. Extremely fine needle

   (Extremely sharp, please use with care. After use, replace the cap and leave it in its original place. Do not discard; otherwise, points will be deducted.)
7. 15 cm ruler
8. One pair of tweezers
9. One 3 mL dropper (to be reused)
10. Anesthetic solution for Xenopus tadpoles.
11. Using anesthetic solution on bullfrog tadpoles
12. P10 pipetteman (micropipettes)
13. P10 straw tip
14. One timer (do not press for timing purposes)
15. One dissecting microscope

Part One: Observation

Question 1: Please draw a diagram to compare the morphological differences between Xenopus laevis and bullfrog tadpoles (5%).

The second part involves experiments.

Experimental steps

[A. Control Group]

1. Place 5 clawed frog tadpoles into a transparent box, then place the transparent box into the black and white outer box, and leave it for 10 seconds.

   Then observe and record the distribution and number of tadpoles in the black and white environments and fill them in Table 1.

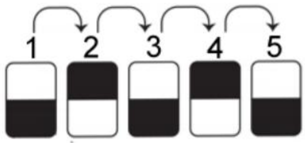
   [This is a repeat of control group one]

2. Take the transparent box out and shake it back and forth and side to side to disturb the tadpoles' distribution.

   Place the transparent box inside the black and white box (with the top and bottom reversed), let it stand for 10 seconds, and then observe.

   Observe and record the distribution and number of tadpoles in black and white environments and fill in the blanks.

   Table 1. [This is a duplicate of control group 2]



3. Repeat step 2 five times (as shown in the diagram on the right).

4. After recording, carefully pick up the tadpoles with a 3mL dropper, put them back into the original culture dish, and cover it for later use.

5. Exchange for 5 bullfrog tadpoles to complete all the above steps.

Control Group Table 1

(The number of tadpoles in the white or black environment is entered repeatedly, and then the percentage is calculated at the end):

Clawed toad tadpoles: white		
One repeat		
Double repeat		
Triple repetition		
Four repetitions		
Five repetitions		
Quantity ratio average%		

Bullfrog tadpoles: white		
One repeat		
Double repeat		
Triple repetition		
Four repetitions		
Five repetitions		
Quantity ratio average%		

Question 2: Please design and create a bar chart to show the response of Xenopus laevis and bullfrog tadpoles to brightness, and use Y...

The axis is represented by % (accounting for 10%).

[B. Eyeball Removal Group]

#This exclusion group experiment focused on surgical techniques and surgical success rate **(20%).**

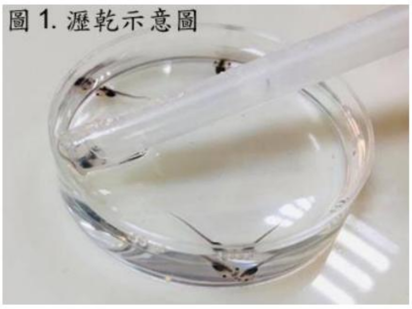


圖 1. 瀝乾示意圖

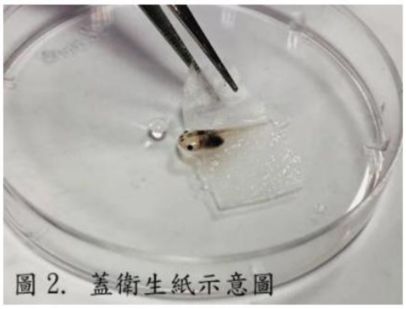


圖 2. 蓋衛生紙示意圖

1. Open the petri dish and aspirate 5 xenopus tadpoles separately, then add more water to the dropper.

After draining the remaining liquid (Figure 1.), transfer it to a culture medium containing anesthetic solution.

In the culture dish, wait 5 seconds after the individual is anesthetized, unconscious, and motionless.

Then suck out the tadpoles, drain them, and place them on top of the original petri dish.

Use tweezers to place a small piece of lint-free paper over each tadpole to secure it.

Position as shown in Figure 2. [Please place with the back facing up and the abdomen facing down.]

2. Eyeball removal: The culture vessel containing 5 anesthetized tadpoles...

The petri dish lid was placed under a microscope, and an eyeball was opened with a needle.

Insert the needle above the edge of the eyeball (Figure 3.) and sever the area behind the eyeball.

After removing the optic nerve, make an incision along the periphery of the eyeball and remove both eyelids.

Side eyeball.

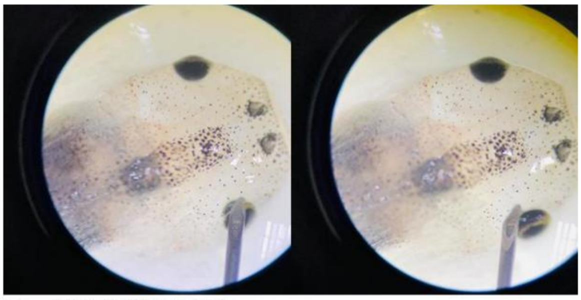


圖 3. 針頭自眼球邊緣上方探入

3. Eye removal: Cover the petri dish containing 5 anesthetized tadpoles.

Placed under a microscope, first use a needle tip to make a circular incision around the eyeball.

The eyeball is then shaved off using a P10 micropipette (Figure **4).**

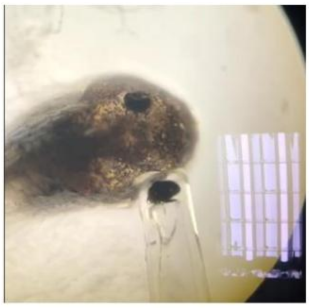


圖 4. 吸取眼球

4. After the surgical procedure, remove the dust-free paper from the tadpoles and add an appropriate amount of [unclear - possibly a specific ingredient or product].

The water is flowing; we await its awakening.

5. After the tadpoles have regained consciousness and are moving normally, place them in a transparent inner box, then place the transparent box inside the black and white outer box. Observe and record the distribution of tadpoles in the black and white environments after 10 seconds (Table 2).

Quantity. [This is a repeat of group one of the ophthalmoectomy.]

6. Take the transparent box out and shake it back and forth and side to side to scatter the tadpoles. Then put the transparent box back in and turn it upside down.

The tadpoles were placed in black and white boxes for 10 seconds, and their distribution in the black and white environments was observed and recorded.

Quantity. [This is the second repetition of the ocular removal group.]

7. Repeat step 6 five times.

8. After recording, return the tadpoles to their original petri dish, cover it, and return it after the exam for evaluation.

Testing skills and surgical success rate.

9. Exchange for **5** more bullfrog tadpoles and then repeat all the above steps.

Eyeball removal group (Table 2)

(The number of tadpoles in the white or black environment is entered repeatedly, and then the percentage is calculated at the end):

Clawed toad tadpoles: white		
One repeat		
Double repeat		
Triple repetition		
Four repetitions		
Five repetitions		
Quantity ratio average%		

Bullfrog tadpoles: white		
One repeat		
Double repeat		
Triple repetition		
Four repetitions		
Five repetitions		
Quantity ratio average%		

Question 3: Please design and create a bar chart to show the response of Xenopus laevis and bullfrog tadpoles to brightness, and use Y...

The axis is represented by % (accounting for 10%).

Question 4: Please compare the differences in their responses to light and suggest possible reasons for these differences (5%).